

Deployment Guide

1. Purpose

The purpose of this guide is to explain how to install, configure, and run the Expense Tracker application on a new development machine.

2. System Requirements

Hardware Requirements

Component Minimum Requirement

Processor	Intel Core i5 or equivalent
RAM	8 GB (16 GB Recommended)
Internet	Required

Software Requirements

Software	Version
Windows	10 / 11
Node.js	v24.x or later
npm	Latest
React Native CLI	Latest
Android Studio	Latest Stable
Android SDK	API 35+
Java JDK	17
Git	Latest
Visual Studio Code	Latest

3. Clone the Project

Clone the repository.

```
git clone https://github.com/haseebmustafa279/ExpenseTracker
```

Move into the project directory.

```
cd ExpenseTracker
```

4. Install Dependencies

Install all required packages.

```
npm install
```

or

```
yarn install
```

5. Android Setup

Install:

- Android Studio
- Android SDK
- Platform Tools
- Build Tools
- Emulator

Verify installation.

```
npx react-native doctor
```

Resolve any reported issues before continuing.

6. Run Metro Server

```
npm start
```

or

`npx react-native start`

7. Run Android Application

Start an emulator or connect a physical Android device.

Run:

`npx react-native run-android`

The application should compile and install automatically.

8. Environment Variables (Future)

Create a `.env` file in the project root.

Example:

`API_BASE_URL=http://localhost:8000/api/`

`JWT_SECRET=xxxxxxxxxxxxxxxxxx`

During production:

`API_BASE_URL=https://yourdomain.com/api/`

9. Backend Deployment (Future Django)

After the Django backend is developed:

Install dependencies.

`pip install -r requirements.txt`

Run database migrations.

`python manage.py migrate`

Create an administrator account.

`python manage.py createsuperuser`

Run the server.

`python manage.py runserver`

Backend URL:

`http://127.0.0.1:8000/`

10. Build APK

Generate a release APK.

`cd android`

`gradlew assembleRelease`

APK Location:

`android/app/build/outputs/apk/release/`

11. Build AAB

Generate an Android App Bundle.

`gradlew bundleRelease`

Output Location:

`android/app/build/outputs/bundle/release/`

12. Git Workflow

Clone

`git clone RepositoryURL`

Pull Latest Changes

`git pull`

Add Changes

`git add .`

Commit

`git commit -m "Implemented Statistics Screen"`

Push

git push origin main

13. Deployment Checklist

Before deployment, verify:

- All screens work correctly.
 - Navigation works correctly.
 - No TypeScript errors.
 - No ESLint warnings.
 - Images load correctly.
 - API endpoints are configured.
 - Environment variables are updated.
 - Release APK builds successfully.
 - Backend server is running.
 - Database migrations have been applied.
-

14. Troubleshooting

Issue	Solution
Metro server not starting	Run <code>npm start --reset-cache</code>
Gradle build failed	Run <code>cd android && gradlew clean</code>
Emulator not detected	Open Android Studio Device Manager
Dependency error	Delete <code>node_modules</code> and run <code>npm install</code>
Cache issue	<code>npx react-native start --reset-cache</code>

15. Production Deployment (Future)

The production deployment process will include:

- Build release APK/AAB.
- Deploy Django backend on a cloud server (e.g., AWS, Azure, or DigitalOcean).
- Configure PostgreSQL database.
- Enable HTTPS.
- Configure domain name.
- Update API_BASE_URL to the production server.
- Publish the Android App Bundle to the Google Play Console.